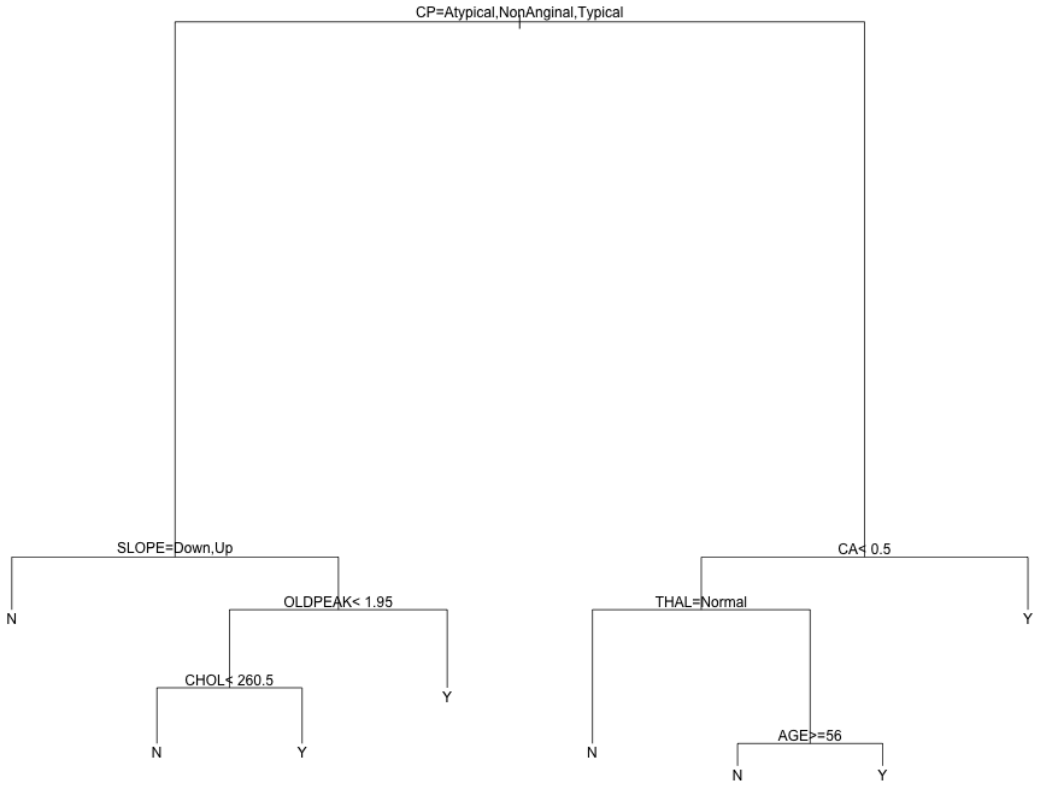
Question 1

1. I think the predictors which are possibly associated with fuel efficiency are Eng.displacement, AspirationSC, AspirationTC, AspirationTS, No.Gears, Lockup.Torque.ConverterY, Drive.SysF because they have p-value < 0.05. Eng.displacement, AspirationTC and Drive.SysF appear to be the strongest predictors of fuel efficiency because they have the lowest p-value, which are significantly lower than the others'.
2. From the result of the Bonferroni procedure, the new significance level is now 0.0056. So the predictors that are associated are changed to: Eng.displacement, AspirationTC, No.Gears, Lockup.Torque.ConverterY and Drive.SysF because they have p-value lower than 0.0056. The AspirationSC and AspirationTS are no longer considered associated with fuel efficiency because their p-value is higher than 0.0056.
3. For every 1 litre increase of engine displacement, the mean fuel efficiency of a car is expected to decrease by 1.331 km/litre, when other predictors remain constant. Besides, the mean fuel efficiency of a car with a front-wheel drive system is expected to be 1.535 km/litre higher than a car without a front-wheel drive system, when all other predictors (characteristics) of both cars are the same.
4. Comb.FE = 16.36119 - 1.31647*Eng.displacement + 0.13369*AspirationOT - 0.57062*AspirationSC - 1.07175*AspirationTC - 1.32489*AspirationTS - 0.17477*No.Gears - 0.57320*Lockup.Torque.ConverterY + 0.19340*Drive.SysA + 1.54754*Drive.SysF - 1.08018*Drive.SysP + 0.28424*Drive.SysR
5.
    a. The predicted mean fuel efficiency for this new car is 13.37209 km/litre. The 95% confidence interval for this prediction is (12.99409, 13.75009) km/litre.
    b. No,my model does not suggest that the new car will have better fuel efficiency than my current car. The confidence interval in 5a is the range of the possible value of the new car's fuel efficiency, my current car's fuel efficiency falls in this interval so statistically, the new car does not significantly have higher fuel efficiency.

Question 2

1. The variables used in the best tree are CP, SLOPE, OLDPEAK, CHOL, CA, THAL, AGE. The best tree has 8 leaves (terminal nodes).
2. Plot:



To predict that someone has heart disease, one of the 4 following conditions has to be met:

1. The chest pain type of the patient is not asymptomatic (it is typical angina, atypical angina or non anginal pain instead), and slope of the patient's peak exercise ST segment is flat, and patient’s exercise induced ST depression relative to rest is greater than or equal to 1.95.
2. The chest pain type of the patient is not asymptomatic (it is typical angina, atypical angina or non anginal pain instead), and slope of the patient's peak exercise ST segment is flat, and patient’s exercise induced ST depression relative to rest is lower than 1.95, and patient’s serum cholesterol is greater than 260.5mg/dl.
3. The chest pain type of the patient is asymptomatic, and the patient's number of major vessels colored by fluoroscopy is greater than or equal to 0.5.
4. The chest pain type of the patient is asymptomatic, and the patient's number of major vessels colored by fluoroscopy is lower than 0.5, and the patient’s thallium scanning results is not ‘normal’ (it is fixed fluid transfer defect or

reversible fluid transfer defect instead), and the patient's age is greater than or equal to 56.

3. According to my tree, the predictor combination that results in the lowest probability of having heart-disease is patient's chest pain type is not asymptomatic (it is typical angina, atypical angina or non anginal pain instead) and slope of the patient's peak exercise ST segment is either down or up.

4. The final model includes SEX, CP, TRESTBPS, CHOL, OLDPEAK, SLOPE, CA. From question 2.1, The variables used by the tree estimated by CV are CP, SLOPE, OLDPEAK, CHOL, CA, THAL, AGE. By comparing them, we can see that the number of variables used in the final (logistic regression) model is equal to the tree estimated by CV's. Both of them use CP, SLOPE, OLDPEAK, CHOL, CA. The final model uses SEX and TRESTBPS while another does not. The tree estimated by CV uses THAL and AGE while the final model does not. The most important predictor in the logistic regression is CP because its 'NonAnginal' level has the lowest p-value.

5. HD = -7.248413 +1.802856*SEXM -2.184817*CPAtypical -2.599144*CPNonAnginal -2.369844*CPTypical +0.021501*TRESTBPS +0.008167*CHOL +0.581819*OLDPEAK +1.931508*SLOPEFlat +0.206602*SLOPEUp +1.074811*CA

6. For one unit increase in CA(number of major vessels colored by fluoroscopy), the log-odds of the patient having heart disease is expected to increase by 1.074811, while other variables in the model stay constant.

7. The step-wise logistic regression model is better than the tree model since its area under the curve is higher than the tree model (0.88 > 0.76). Besides, the step-wise logistic regression has higher classification accuracy than the tree model (0.83 > 0.78). These mean that the step-wise logistic regression model's ability to distinguish between positive and negative cases is much better. Plus, it also has significantly lower logarithmic loss than the tree model (39.44 < 460.52) meaning that it is much less prone to be confidently wrong.

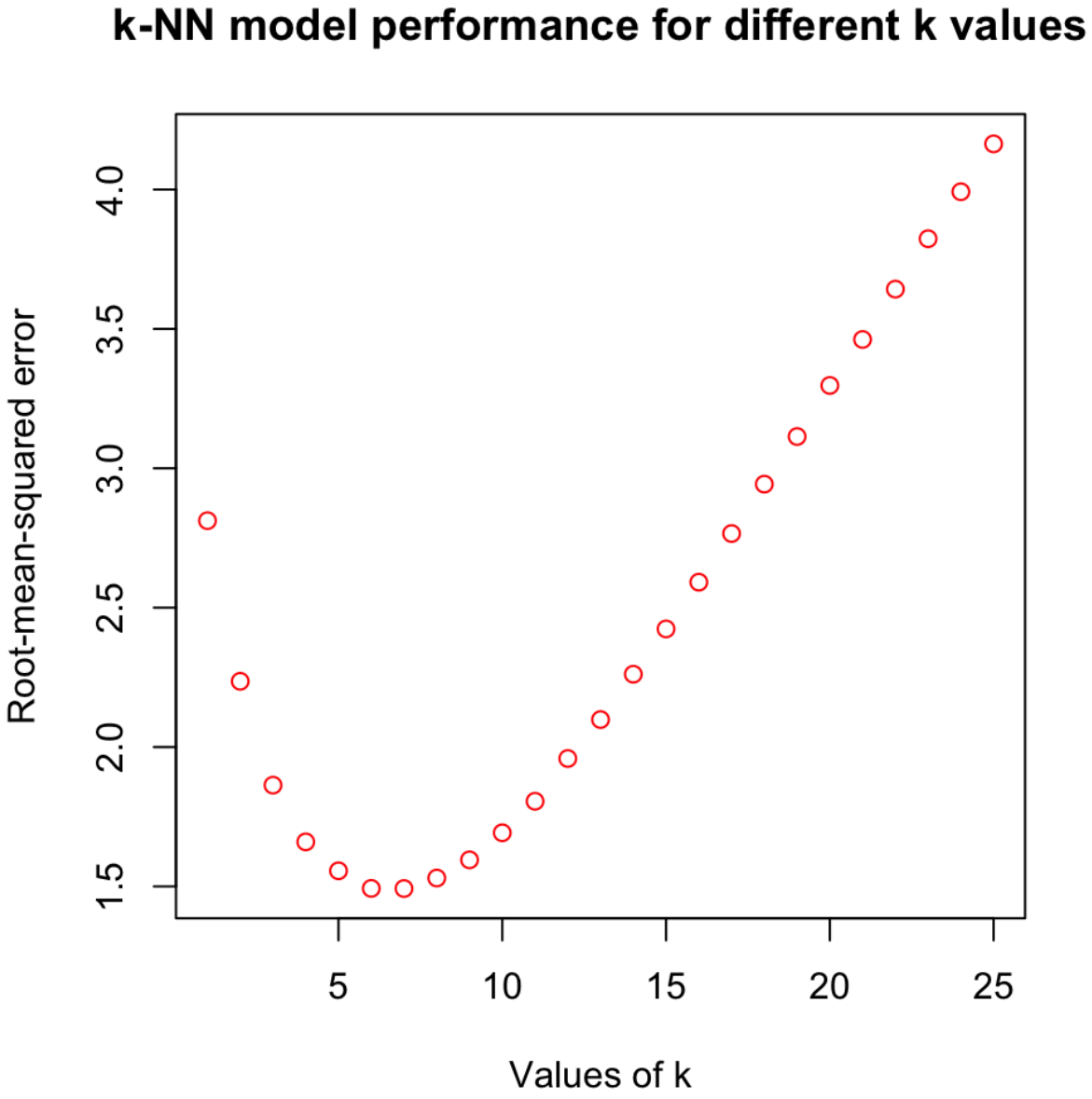
8.a. The odds of having heart disease for the 10th patient in the test dataset calculated by the tree model found using cross-validation 13.25.

8.b. The odds of having heart disease for the 10th patient in the test dataset calculated by the step-wise logistic regression model is 1993.977.
The predicted odds for the two models both indicate that the 10th patient is at high risk of having heart disease because both of the models' odds are much larger than 1.

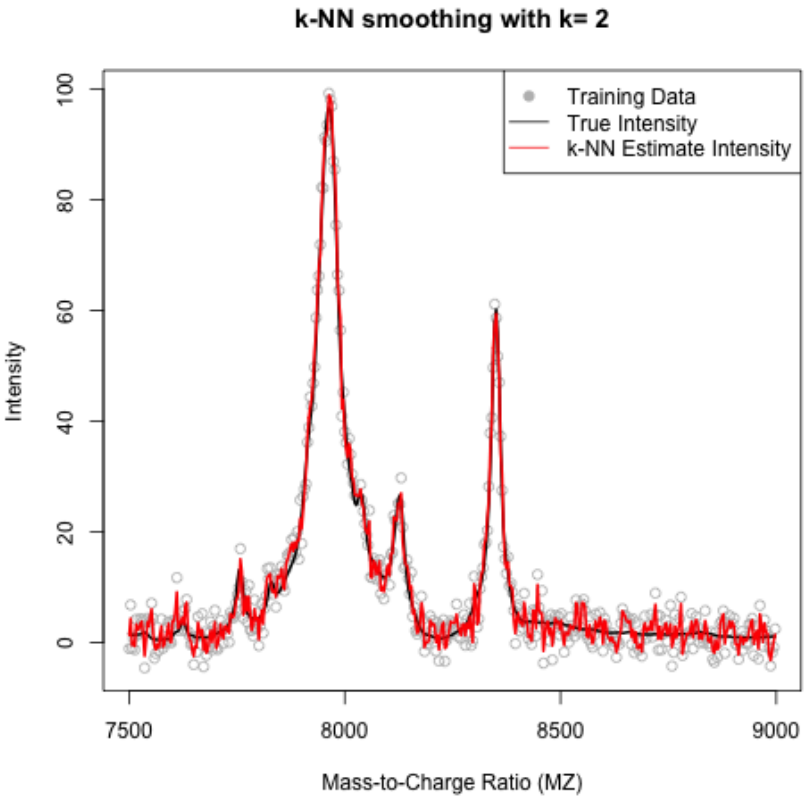
9.I directly bootstrap the difference in odds of patient 65 and patient 66 and the 95% confidence interval is (4.68, 116.67) . This suggests that there is a real difference in the population odds of having heart disease between these two individuals because the interval is always positive and does not contain 0.

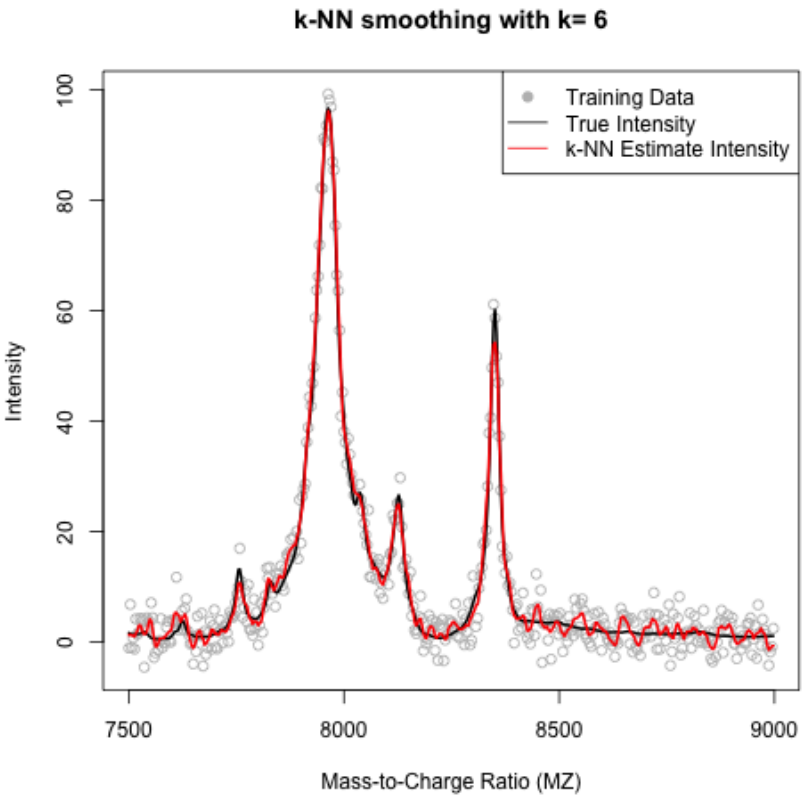
Question 3

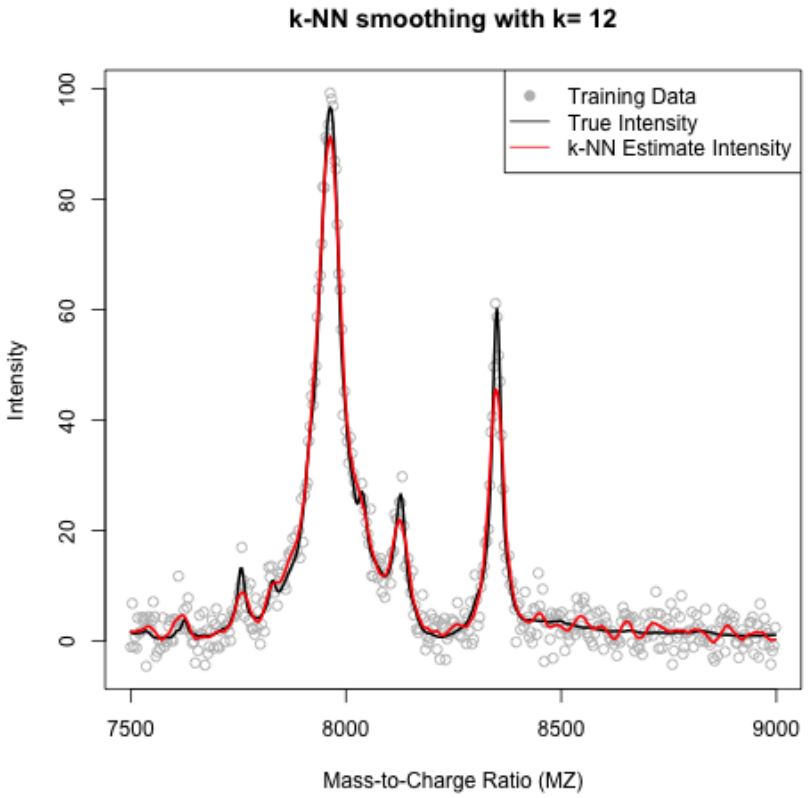
1.

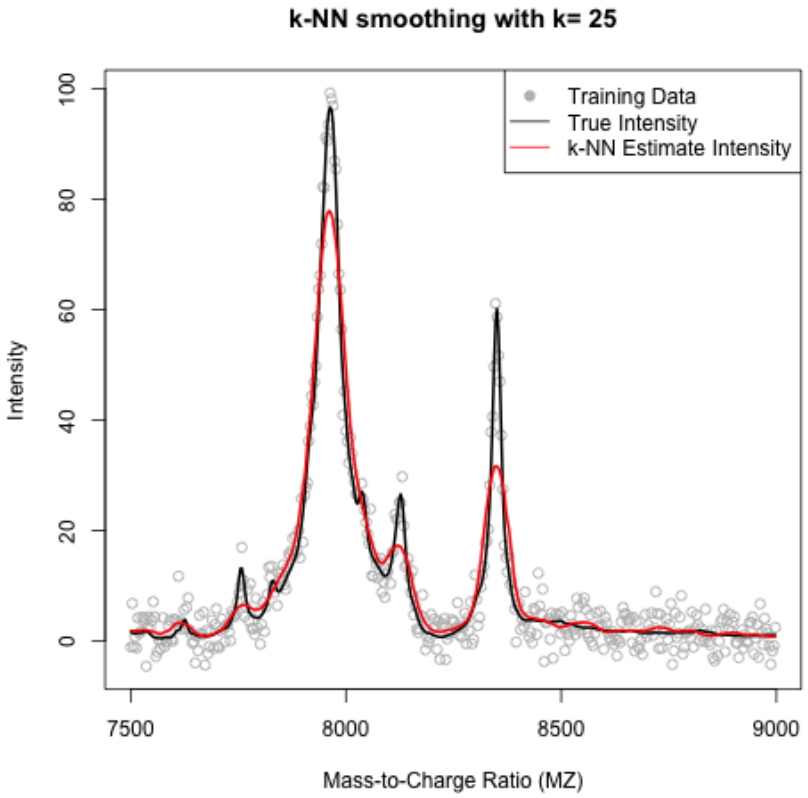


2.


k-NN smoothing with k= 2
Training Data
True Intensity
k-NN Estimate Intensity
Intensity
0
20
40
60
80
100
7500
8000
8500
9000
Mass-to-Charge Ratio (MZ)


k-NN smoothing with k= 6
Training Data
True Intensity
k-NN Estimate Intensity
Intensity
0
20
40
60
80
100
7500
8000
8500
9000
Mass-to-Charge Ratio (MZ)





3. As k increases, the model transitions from increasing its sensitivity that tends to learn noise and leads to high variance, to being overly simple that leads to high bias. As we can see from the 4 plots in question 3.2, in graph k=2, the red line(the

estimate of spectrum) is spiky and does not exactly lean on the black line because it is too flexible that it learns the random noise and causes overfitting. In graph k=6, we can see the red line fits closer to the black line (less spiky), this is the most optimal model in these 4 different k-value models. In graph k=12, the red line has become too smooth that the model now brings high bias and misses important features. In graph k=25, the red line has missed out most of the black line because it is being too inflexible and causes a very high bias.
Besides, from my plot of question 3.1, we can see that low k (around <6) has high root-mean-squared error(RMSE) because it has high variance as I just mentioned. When k is 6 or 7, the RMSE reaches its lowest value, meaning that the estimate is the most accurate, and the bias and variance have a good balance now. When k is greater than 7, RMSE tends to grow larger because of high bias as I just mentioned.

4. The method selects k = 7, which is identical to the value of k that would minimise the actual mean-squared error (in Question 3.1 is also k = 7. This has demonstrated the power of cross validation.
5. The estimate of the standard deviation of the sensor/measurement noise that has corrupted our intensity measurements is 2.5180.
6. The estimated spectra for k = 6 plotted in Q3.2 achieve our dual aims of providing a smooth, low-noise estimate of background level as well as accurate estimation of the heights of the peaks. The reason is that when the true spectrum is flat, the estimated line is smooth enough to lie around the true line because it averages out the random noises. When the true line is at peak, the estimate line is flexible enough to capture the true line (even at the highest point).
7. From the smoothed signal produced using the value of k found in Q3.4, the value of MZ corresponds to the maximum estimated intensity is 7963.3.
8. The 95% confidence interval using the k = 7 (determined in Q3.4) :(90.00, 98.97) .
   The 95% confidence interval using the k = 3: (92.12, 99.23).
   The 95% confidence interval using the k = 20: (70.60, 93.01).
   I think these confidence intervals vary in size as we change the value of k because we are estimating the intensity value (which is actually at **peak**), the lowest k (k= 3) model uses the most localized neighbourhood so it gives the most accurate result and the smallest width of confidence interval (99.23-92.12= 7.11). When k increases (to k=7), the neighbourhood is enlarged, hence more variability and wider confidence interval. When k = 20, the large neighbourhood brings high bias, leading to the widest confidence interval.